

MA3236 NONLINEAR PROGRAMMING

Semester 1, AY 2026/2027

Tutorial 4: One-dimensional methods and Newton methods

Review: Lecture 8

- **Iteration and stopping.** On an open domain $\mathcal{D} \subseteq \mathbf{R}^n$, initialize, update, and test:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)} \in \mathcal{D}.$$

Typical tests use a small gradient norm, a small change in iterates or function values, or a resource limit. For example,

$$\|\nabla f(\mathbf{x}^{(k)})\| \leq \epsilon_g \quad \text{or} \quad \|\nabla f(\mathbf{x}^{(k)})\| \leq \epsilon_g \max\{1, \|\nabla f(\mathbf{x}^{(0)})\|\}.$$

A small step may signal stagnation; reaching the iteration limit is not a convergence certificate. If f^* is known, an objective-gap test is also possible.

- **Restricting to a line.** For a fixed direction, define

$$\phi^{(k)}(\alpha) = f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}), \quad \mathcal{I}^{(k)} = \{\alpha \geq 0 : \mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)} \in \mathcal{D}\}.$$

Exact line search minimizes $\phi^{(k)}$ over $\mathcal{I}^{(k)}$, when the minimum is attained. Solving $(\phi^{(k)})' = 0$ alone need not find that minimum.

- **Rates and cost.** For $e^{(k)} = \|\mathbf{x}^{(k)} - \mathbf{x}^*\|$,

$$\begin{array}{l|l} \begin{array}{l} Q\text{-linear} \\ Q\text{-superlinear} \\ Q\text{-quadratic} \end{array} & \begin{array}{l} e^{(k+1)} \leq ce^{(k)} \text{ eventually, } 0 < c < 1 \\ e^{(k+1)}/e^{(k)} \rightarrow 0 \\ e^{(k+1)} \leq C(e^{(k)})^2 \text{ eventually.} \end{array} \end{array}$$

Iteration count, derivative evaluations, linear solves and storage are different costs. A local rate is not a guarantee from every initial point.

- **Three scalar methods.** Write $\ell^{(k)} = b^{(k)} - a^{(k)}$ and $\rho = (\sqrt{5} - 1)/2$.

Method	Assumption and update	Main cost / rate
Bisection on f'	Continuous f' and a sign-changing bracket; test its midpoint and retain the sign-changing half.	One new f' per reduction; $\ell^{(k+1)} = \ell^{(k)}/2$.
Newton on f'	$x^{(k+1)} = x^{(k)} - f'(x^{(k)})/f''(x^{(k)})$; the denominator must be nonzero.	One f' and f'' per update; locally Q-quadratic under regularity.
Golden section	Strict unimodality on a closed bracket; compare $f(\lambda^{(k)})$, $f(\mu^{(k)})$, where $\lambda^{(k)} = b^{(k)} - \rho\ell^{(k)}$, $\mu^{(k)} = a^{(k)} + \rho\ell^{(k)}$.	Two initial values, then one new value per comparison; $\ell^{(k+1)} = \rho\ell^{(k)}$.

For either bracketing method, the final midpoint error is at most $\ell^{(k)}/2$. This bound does not assert a fixed ratio of actual errors.

Review: Lecture 9

Set $\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)})$ and $\mathbf{H}^{(k)} = \mathbf{H}_f(\mathbf{x}^{(k)})$.

- **Pure Newton: solve the model's stationarity equation.**

$$m^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{H}^{(k)} \mathbf{d}, \quad \mathbf{H}^{(k)} \mathbf{d}_N^{(k)} = -\mathbf{g}^{(k)}.$$

Take the full step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{d}_N^{(k)}$. A nonsingular Hessian gives a unique solution of the equation, not necessarily a minimizer of the model or of f .

- **Local quadratic convergence.** At a stationary point $\mathbf{x}^*$, assume $\mathbf{H}_f(\mathbf{x}^*)$ is non-singular and $\mathbf{H}_f$ is locally Lipschitz. Pure Newton converges locally Q-quadratically when $\mathbf{x}^{(0)}$ is sufficiently close. Positive definiteness of $\mathbf{H}_f(\mathbf{x}^*)$ is not required: the target can be a saddle point.
- **Descent and shifting.** A descent direction satisfies $(\mathbf{g}^{(k)})^T \mathbf{d}^{(k)} < 0$ and decreases f for sufficiently small positive steps. Choose $\tau^{(k)} \geq 0$ so that

$$\mathbf{B}^{(k)} = \mathbf{H}^{(k)} + \tau^{(k)} I \succ 0, \quad \mathbf{B}^{(k)} \mathbf{d}_M^{(k)} = -\mathbf{g}^{(k)}.$$

Then $(\mathbf{g}^{(k)})^T \mathbf{d}_M^{(k)} < 0$ if $\mathbf{g}^{(k)} \neq \mathbf{0}$. The implementation tests, e.g., $\tau = 0, 0.1, 0.2, 0.4, \dots$ and uses the first positive-definite matrix, followed by a full step.

- **Positive-definite model $\neq$ majorization.** The modified model is

$$M^{(k)}(\mathbf{d}) = f(\mathbf{x}^{(k)}) + (\mathbf{g}^{(k)})^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \mathbf{B}^{(k)} \mathbf{d}.$$

If $M^{(k)}(\mathbf{d}) \geq f(\mathbf{x}^{(k)} + \mathbf{d})$ on the relevant region, minimizing it guarantees decrease. Positive definiteness alone does not imply this inequality: pure and shifted full steps can both increase f .

- **Armijo backtracking line search.** For a descent direction, start with $\alpha = 1$ and multiply it by $\beta \in (0, 1)$ until the trial point belongs to $\mathcal{D}$ and

$$f(\mathbf{x}^{(k)} + \alpha \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + \sigma \alpha (\mathbf{g}^{(k)})^T \mathbf{d}^{(k)}, \quad 0 < \sigma < 1.$$

Backtracking then terminates in finitely many trials. Newton–Armijo uses the raw Newton direction only if it is well defined and descending; otherwise it uses a shifted safeguard.

- **What line search does and does not promise.** It enforces sufficient decrease, not a universal global-minimizer guarantee. Pure Newton may converge without it. Near a regular local minimizer, the unit Newton step is eventually accepted when $0 < \sigma < 1/2$ and the local smoothness assumptions hold.

1. **Three one-dimensional methods.** Let $\varphi(t) = t^4 + 2t$ and $\psi(t) = t^2 + 2t$, both on $\mathcal{D} = \mathbf{R}$.

- (a) For φ , carry out two bisection reductions on φ' from $[-1, 1]$. How many reductions in total guarantee a midpoint error of at most 10^{-3} ?
- (b) For φ , compute two pure Newton updates from $t^{(0)} = -1$ exactly, without early stopping.
- (c) For ψ , carry out two golden-section reductions from $[-1, 1]$. Show the interior points, function-value comparisons and retained brackets, reusing a computed value when possible.

2. **Newton and exact line search on a quadratic.** Consider

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathcal{D} = \mathbf{R}^2,$$

with $\mathbf{x}^{(0)} = (1, 1)^T$.

- (a) Compute the pure Newton direction and the next iterate. Why does this full step solve the minimization problem?
- (b) For a positive-definite quadratic and a descent direction $\mathbf{d}$, derive

$$\alpha^* = -\frac{\nabla q(\mathbf{x})^T \mathbf{d}}{\mathbf{d}^T \mathbf{Q} \mathbf{d}}$$

for exact line search over $\alpha \geq 0$. Apply it at $\mathbf{x}^{(0)}$ with $\mathbf{d}^{(0)} = (-1, -1)^T$: find the step and next point. Does this line search reach the global minimizer of q ?

3. **Shifting is not a step-length safeguard.** Use $\varphi(t) = t^4 + 2t$ from Question 1, now at $t^{(0)} = 0$.

- (a) Can pure Newton take a step? For shifted Newton, test $\tau = 0, 0.1, 0.2, 0.4, \dots$ and select the first positive scalar $B^{(0)} = \varphi''(t^{(0)}) + \tau$. Compute

$$d_M^{(0)} = -\frac{\varphi'(t^{(0)})}{B^{(0)}} \quad \text{and} \quad \varphi'(t^{(0)})d_M^{(0)}.$$

- (b) Compute the full shifted step and its function value. Does the positive-definite shifted model majorize φ at that trial point?
- (c) Apply Newton–Armijo with this shifted direction, $\bar{\alpha} = 1$, $\beta = 1/2$ and $\sigma = 10^{-4}$. List every trial step length up to acceptance, and give the accepted iterate. Explain which failure shifting fixes and which backtracking fixes.

4. **An exact Newton sequence and a stopping index.** Consider

$$f(x_1, x_2) = x_1^4 + 2x_1^2x_2^2 + x_2^4, \quad \mathcal{D} = \mathbf{R}^2,$$

with $\mathbf{x}^{(0)} = (a, a)^T$, where $a \neq 0$.

- (a) Show that every full Newton update is well defined and

$$\mathbf{x}^{(k)} = \left(\frac{2}{3}\right)^k (a, a)^T.$$

- (b) Identify the limit and the convergence rate of the iterates. Does this contradict the local quadratic convergence theorem?
- (c) For $a = 1$, find the exact ratio $f(\mathbf{x}^{(k+1)})/f(\mathbf{x}^{(k)})$ and the smallest integer k for which $f(\mathbf{x}^{(k)}) \leq 10^{-6}$.